

Energy-sharing mechanisms for energy community members under different asset ownership schemes and user demand profiles

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ABSTRACT

Renewable Energy Communities are aggregations of final users/prosumers whose energy comes from renewable power generators. This work systematically assesses the rules of the current and new approaches to sharing the generated benefits obtainable from renewable energy production and its consumption among members of a virtual net-metering energy community. In this work, three sharing mechanism algorithms are proposed to distribute the energy community net profit among the members, utilizing a Token currency, in proportion to the service they provide: financing service, self-consumption service, and both financing and self-consumption services. An energy community, comprising of 100 households and supplied by a 100 kWp photovoltaic system, has been simulated. Six scenarios were designed to determine how profits could be shared between users and prosumers. We focused on the impact that the adoption of a specific sharing mechanism could have on the different categories of members (e.g., low-energy users, high-energy users with ownership, etc.).

Our results show that members with a low-energy demand, and who are poorly synchronized with the renewable plant load, would benefit the most from a sharing algorithm that equally rewards community members. On the other hand, high-energy load members who are well synchronized would benefit the most from a sharing strategy that rewards the actual contribution of the user to the realization of the overall shared energy. As far as ownership of the renewable plant is concerned, users would benefit from owning the renewable plant when their load has a good temporal match with the renewable power supply.

1. Introduction

Distributed power generation, based on variable renewable energy sources, is spreading throughout the world. In the building sector, rooftop or building-integrated photovoltaic (PV) systems [1] are finding renewed employment, transforming final users into *prosumers* who are willing to own and install distributed renewable energy technologies, storage systems, and demand response infrastructures [2]. Besides

distributed generation, a trend that is supported by EU policies is emerging which aggregates the demand of final users via *energy communities* (EC). According to RESCooP, the European federation of citizen energy cooperatives: “community energy is any project or initiative where people have ownership or a meaningful say in the running of renewable energy or energy related services. We also consider energy efficiency projects where the community is involved in collectively reducing their energy use as important community energy” [3].

In Europe, in 2019, approximately 50% of the population lived in apartments (multi-family residential buildings), with an annual tendency of this share increasing [4]. The EU *renewable energy directive* (RED II¹) has recently introduced the category of *jointly acting renewable self-consumers*, i.e., *collective self-consumers* (CSC) (not necessarily residential users) who are supplied by a production unit that is installed in their premises, e.g., a condominium with a rooftop PV installation shared among the building users. The final users can also aggregate and share energy with other buildings, thus forming a second category called ‘renewable energy community’ (REC). A proximity

Abbreviations: EC, Energy Community; EU, European Union; FiP, Feed-in-Premium; FiT, Feed-in-Tariff; GIS, Geographic information system; NPV, Net Present Value; OF, Ownership-Financing mechanism; P2P, Peer-to-peer; PBT, Payback time; PCC, Pearson correlation coefficient; POD, Point-of-delivery; PV, Photovoltaic system; REC, Renewable energy community; RES, Renewable Energy Sources; SE, Shared Energy; SM, Sharing Mechanism; TC, Token Currency; ToU, Time-of-Use; UT, User Type.

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¹ Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources.

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criterion is generally put in place. The energy can be shared among final users who are connected at the same electric aggregation levels (e.g., *medium-voltage/low-voltage* MV/LV transformation cabin) or according to their postal code or another proximity criterion.

Regardless of which configuration is adopted (CSC or REC), the open question is how to share the generated benefits from the self-produced energy and through which internal distribution algorithm (sharing mechanism). In this regard, it has become relevant to assess how to share energy community² profits among final users or building units, considering different types of distributed generation asset ownership.

There is a lack of previous work on sharing energy among different users since energy communities have only recently been introduced into the energy policy landscape through the adoption of Directive RED II, which became effective at the end of 2018. Many member countries are still finalizing the transposition of this directive into national laws (see, for instance, Ref. [5]). Previous work mostly focused on a specific topic, that is, peer-to-peer energy trading. However, this approach is quite different from how an energy community works in the real world. An energy community does not (necessarily) need a rather complex digital framework for a direct exchange of the commodity (e.g., renewable electricity) among users. The distribution system operator (DSO) is likely to remain the only cyber-physical infrastructure through which electricity and information will flow and be shared among community members. Of course, a digital upgrade of the distribution network is often required for a quasi-real-time accounting (e.g., at least hourly) of the exchanged energy flows. Then, energy sharing among members becomes essentially a matter of a virtual exchange rather than an actual exchange since the distribution grid is always there to connect both users and prosumers, regardless their membership to specific energy community.

In this context, we have explored the mechanisms through which users can decide to share the benefits derived from collective self-generation of renewable electricity and collective self-consumption. To the best of our knowledge, there is no prevailing practice, even among the first real case studies of RECs, on how to share the profits. Hence, we would here like to provide a formal framework to 1) propose relevant sharing mechanisms that could be applied in the real world, and 2) understand what mechanism can provide benefits to a given type of user/prosumer.

In the coming years, there will probably be a wide proliferation of RECs, at least in the European area. We believe it is important to establish governing principles for a fair remuneration of the involved members.

This work focuses on an electricity-based energy community powered by *renewable energy sources* (RES) and operated through a virtual net-metering scheme, see Fig. 1. The aim is to systematically investigate how sharing mechanisms impact different types of EC members. To the best of our knowledge, this topic has not yet been investigated either extensively or systematically in literature. The energy community business model comprises a two-level trading system. At the first level, revenues are generated from the produced energy and the incentives on the *shared energy* (SE). The EC manager receives these revenues. At the second level, the EC manager distributes the revenues to the EC members according to the chosen *sharing mechanism* (SM). Three intuitive and straightforward sharing mechanisms have been proposed in this work, whose algorithms can easily be implemented and require very little computational power. The sharing mechanisms were designed to reward a different energy service provided by one of the members to the EC: a financing service, a self-consumption service, or both. The adopted

investigation approach involved simulating a synthetic energy community case study, with its generation assets, the load demand of the EC members, and the energy exchanged with the electric distribution grid. A set of six EC scenarios was then developed to analyze what sharing mechanism offers the most benefit to the different categories of users, and through what generation asset ownership scheme and financing mechanisms. The novelty of the approach is that it differentiates the EC members based on their (hourly) synchronism between load demand and RES generation, which has a major impact on the EC business model and repercussions on the allocation of profits among EC members. This work is structured as follows. Section 2, the literature review, outlines the boundaries of this work, describes what energy communities are and what characteristics they exhibit under specific energy framework and policy conditions. Emphasis has been placed on the current EU legislative framework and the prevailing support mechanisms in different EU member states. Section 3 introduces the simulation framework that has been used to model the energy community: a generic business model, based on shared energy incentives, the mathematical formulation of the three different sharing mechanisms, and the scenarios designed to systematically investigate the impact of the sharing mechanism on different types of EC members. The simulation results are analyzed and discussed in Section 4. Finally, Section 5 offers the conclusions.

2. Literature review

The literature review has been organized to offer an understanding of the context of this work, that is, to define the objective of the investigation, the methodological approach, and the gap this research work attempts to fill. Section 2.1 presents how the term “energy community” is used in literature to indicate different concepts. The two main enabling policy frameworks for an energy community are reviewed in 2.2 and how different countries are implementing them. Section 2.3 reviews the methodologies used most frequently to perform theoretical investigations of energy communities. Finally, a review of the sharing mechanisms is presented.

2.1. Energy communities

The idea of energy community is inherently linked to the possibility of different users collectively exploiting the energy generated by a power plant, for example renewable energy. Self-consumption of RES energy is a topical subject on electricity markets. Moura and Brito highlighted how, until now, the rollout of self-consumption systems has mostly been conducted on a ‘one-to-one’ basis (one production system per consumption point) [6]. This paradigm inherently limits the diffusion of roof-top PV systems in cities, since the condominium building type is better suited to a ‘one-to-many’ configuration (one production system whose energy is shared among the occupants of the same building). Schiera et al. showed, through a GIS-based agent-based modeling approach, that the diffusion of rooftop PV systems in a city district is boosted when the regulatory framework allows the one-to-many configuration [7]. Until 2019, i.e., when the RED II directive was enacted, many EU member states did not allow such a configuration. Now, the opportunity of aggregating end-users is becoming a reality, thanks to the CSC and REC configurations enabled by the RED II directive.

In recent years, various idea of prosumers aggregation have been investigated and conceptualized [8]: low carbon communities [9], sustainable communities [10], renewable energy communities [11], community microgrids [12], sustainable energy communities [13], clean energy communities [14], community energy [15], renewable energy cooperatives [16] civic energy communities [17] and renewable energy clusters [18] to name but a few.

The different names given to these user aggregations reflect a specific connotation or perspective of the authors. Energy communities were classified by Moroni et al. [19] considering the EC purpose and location,

² Shared energy corresponds to the amount of generated renewable energy that is consumed by the energy community members within a time slot of fixed duration. For instance, if the calculation of the shared energy is done on an hourly basis, shared energy corresponds to the minimum between the generated energy and the collective load of the EC members.

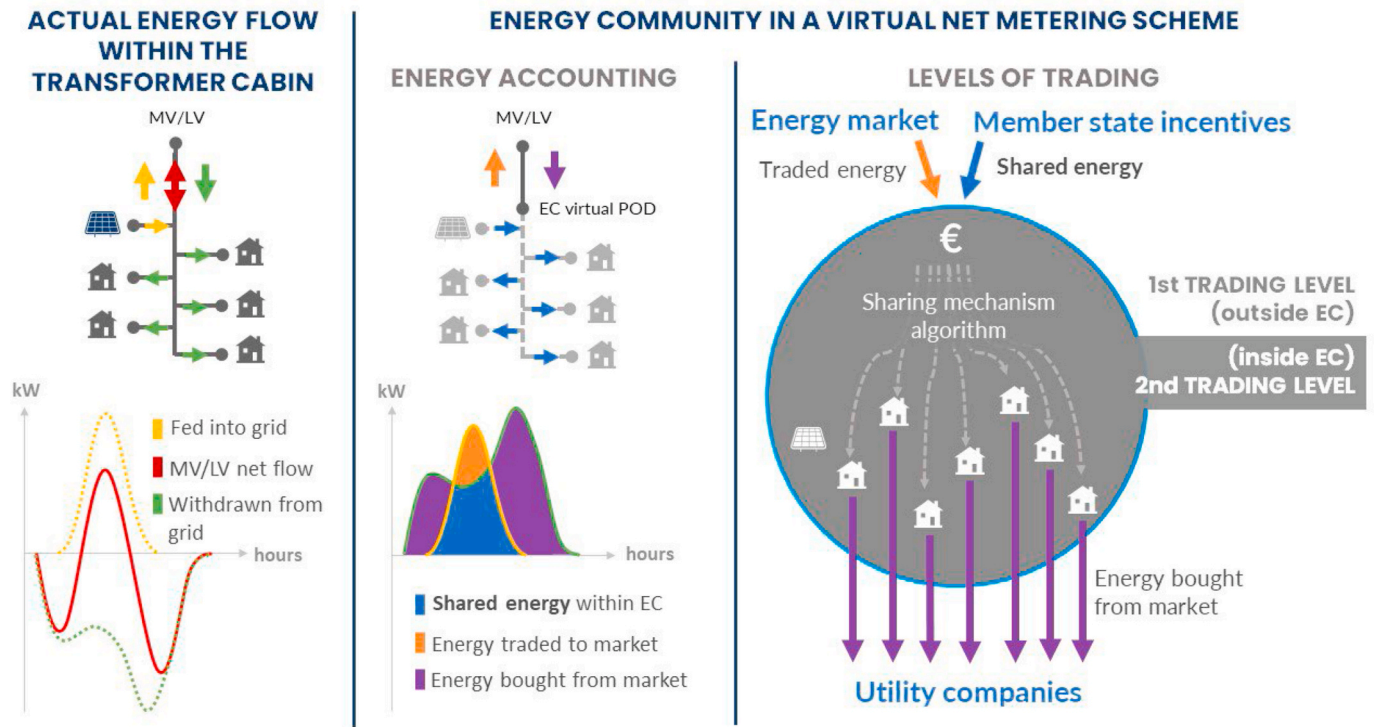


Fig. 1. Physical flow of energy (left) within an energy community Vs. its calculation in a virtual net-metering scheme (center); schematization of the two-level trading system (right).

and discriminating between energy-based and multi-purpose EC, as well as the distribution of the community at different spatial aggregation levels. Gui et al. [14] divided EC into centralized, distributed, and decentralized, distinguishing member's identity and goal cohesion on a spatial level. Kairala et al. [8] compared EC with respect to their energy system integration and the provided energy services. Moreover, energy communities can exhibit different topologies, according to the users' connection with the public grid [20]. Here, the main distinction is between EC connected directly to the public grid and those "behind the meter", i.e., users who are connected to a private grid but possibly with the same *point-of-delivery* (POD) on the public grid [21], which, in literature, is often referred to as a microgrid [22]. In this work, user aggregations have generically been labeled as *energy communities*.

2.2. Overview of value generation schemes in a renewable energy community

The aim of this section is to clarify the context of this work by offering a broad overview of the prevailing instruments through which renewable energy communities can operate and guarantee their economic sustainability. First, the current energy policies and supporting mechanisms of different countries were reviewed. We then defined the research domain of this study to represent today's most recurrent policies and supporting schemes of renewable energy community projects.

The energy community configurations mentioned in Section 2.1 require specific regulatory frameworks to be put in place. The key aspect is how self-generated energy is shared and exchanged among community members and with respect to the distribution grid. The two main schemes proposed in the literature are *peer-to-peer* (P2P) and *virtual metering* [6]:

a) The peer-to-peer scheme allows self-produced energy to be traded among different points-of-delivery, or with the wholesale market, and the exchanged energy volume and price to be defined autonomously [23].

b) The virtual metering scheme considers the net energy flux exchanged between different points-of-delivery, i.e., users, as if they had the same virtual POD. Thus, an EC has the same virtual POD, even though the members are physically connected to different points of the grid [24].

The energy transaction between two points-of-delivery is a complex procedure that involves coordinating various stakeholders with different roles in the process (TSO, DSO, energy provider, retailers, prosumers, consumers). It requires a legal and technical framework to harmonize the whole process, define a shared protocol on how energy is measured, and to share information with the involved parties and store it in a proper ledger [25].

An energy policy aiming at regulating the REC should define the internal billing mechanism, regulate the payment of the grid charges, define and adopt specific incentives (e.g., premium tariffs) for the REC, specify the eventual taxation of revenues, and finally the mechanism to remunerate the appointed parties for the energy withdrawn or fed into the grid. Grid usage is subjected to payment of local network charges that are used to sustain the operational and maintenance-related costs. The typical procedure used to collect network tariffs is to include them in the end-user electricity bill charged by the energy retailer. The collected amounts are then transferred to the respective grid operators [26]. For example, on the electric market, end-users connected to a low-voltage distribution grid usually pay for the entire voltage energy transfer, while high-voltage consumers do not pay for any low-voltage network fee. An energy community that exchanges and self-consumes self-generated electricity at the same voltage level among their members could thus avoid network losses (e.g., transformation and distribution/transmission losses) [27]. Therefore, the shared energy among EC members collocated within the same voltage level cabin should translate in lower network charges, in agreement with the direction of having more cost-reflective energy tariffs [28]. The billing mechanism in a traditional power system is managed by a central body, which acts like an energy provider rather than a national agency. A centralized billing

scheme implies that the users' energy bills are not managed directly by the energy community, unless the EC itself is the appointed energy provider. Many authors are investigating decentralized billing mechanisms based on distributed ledger technologies. The blockchain architecture is one of such mechanisms [29]. Distributed ledger technologies enable P2P transactions, without the mediation of any third-party, such as an energy provider. Zia et al. [30] have recently reviewed the proposed distributed ledger architectures.

Energy policies define not only the architecture of the billing mechanism, but also how energy procurement is accomplished with an energy community to satisfy its internal energy demand. The most common energy procurement strategies are:

- Buying energy from an energy utility: the most common contracts are those at a fixed price or with *time-of-use* (ToU) tariffs;
- Collective purchase group contracts that achieve volume discounts from the energy utility: this type of final user aggregation is aimed at bargaining a volume discount contract with the utilities;
- Buying energy from the wholesale market, which is only allowed for market operators; therefore, the EC would have to become a market operator to make use of this strategy;
- Buying energy directly from an energy producing company through a *power purchase agreement* (PPA): PPAs are long-term contracts between a provider and a purchaser (even aggregation of the purchaser) who agree on all the terms for the selling of energy, including the volume, payment terms, source type, delivery schedule, etc. On the other hand, different energy policies have been implemented by different countries to remunerate the energy generated (and fed to the grid) from selected energy sources, most often renewables:
- Bidding on the wholesale energy market, although a market operator license is needed to operate on the market;
- A bilateral contract or PPA: this is often used to finance generation assets independently, as it guarantees a fixed energy price for a long-term period, thereby reducing the financial risk;
- Feed-in-tariff (FiT): this is a fixed price that is set by the national authority to incentivize the energy fed into the grid. The price is often above the market price and may depend on the plan size, location, and technology [24].
- Feed-in-premium (FiP): unlike the FiT tariff, this pays for all the generated energy instead of just the energy fed into the grid;
- Net purchase and sale: this is a mechanism by which the utilities or grid companies buy the energy fed into the grid at a set price. Unlike FiT, the price is close to the average whole market price [31];
- Net-metering/Virtual net-metering: this is a mechanism by which the withdrawn energy is reduced by the amount of energy fed into the grid within a certain time frame [31]. The time window within which the net-metering count is carried out depends on the technical regulations of the network distributor; a monthly billing period is generally used, but, in principle, it could also be yearly, hourly, or even quarter-hourly. Since the retail energy price is higher than the average whole market price, the economic benefit of the net-metering scheme is higher than the *net purchase and sales revenues* for the same amount of energy fed into the grid;
- Virtual self-consumption: the withdrawn energy is paid on a retail basis, while the self-consumed energy is incentivized at a price that is set by the national authorities [24].

Many countries are approving specific energy policies to support EC aggregations by allowing certain combinations of energy procurement schemes and energy generation incentives. Zhang et al. [32] reported the recent developments of P2P pilot projects around the world. At the same time, virtual metering energy policies have reached a mature state of development in many countries [6] and are now widespread through European countries, thanks to a common legislative framework on energy communities [33,34].

The status of the evolving regulatory framework on collective

renewable energy prosumers in the EU was summarized in the work of Inès et al. [35]. Since 2018, many EU countries, such as Greece, Portugal, Belgium, Lithuania, France, Austria, The Czech Republic, Luxembourg, Estonia, Ireland, Italy, Sweden, Spain, and Slovenia, have implemented specific actions to adopt the EU directive framework either fully or partially, as summarized in Table 1, which is taken from the data of Frieden et al., [36]. These authors reported on the transposition of the EU directives in EU members states until December 2020.

The National Renewable Energy Laboratory (NREL) reported that many states in the United States of America had implemented different legislation and incentive mechanisms to support the virtual metering framework that enables energy community projects [37]. However, to the best of our knowledge, there are not many complete and detailed reports on the European Union, due to the ongoing evolution of the UE member state legislation [38]. Table 2 provides an overview of those EU member states where energy community support mechanisms are already in place. However, Table 2 does not report any self-consumption, single-building incentive mechanisms or behind-the-meter policies, e.g., for such countries as Luxembourg [36]. The United Kingdom is reported as an exception because, in 2015, the government, after many years of FiT subsidies to support energy community initiatives, decided to close this scheme for PV and wind projects, thus resulting in a drastic downscaling of these EC projects [39].

2.3. Modeling energy communities

Simulations and modeling techniques are system analyses that are used to provide quantitative data to support the setting up of projects, energy strategy developments, decision-making processes, and policy design [47]. The main objectives, which can be subdivided into several often overlapping categories, are [48]:

- operational decision support, which is used to optimize the operation, management, and dispatching of a technology portfolio;
- power system and power flow analysis, which are aimed at determining dynamic issues and network faults;
- investment decision support, which is used to determine the type and capacity of the technology mix that maximizes the system goals;
- scenario planning, which is used to investigate alternative development scenarios according to endogenous (i.e., energy demand, diffusion, etc.) and exogenous (i.e., energy policies, technology costs, etc.) factors.

A recent literature review by Gjorgievski et al. [71] shows the most common methods, modeling objectives, and constraints used to investigate and design energy communities. Modeling an energy community means investigating its design and configurations, in terms of technology, structure, and organization within a technical and policy framework, with the aim of maximizing the EC and its members' utility. The goal of an energy community and the desired impacts are formulated in a business model and then translated into an optimization problem rather than a heuristic or metaheuristic simulation. Both approaches are widely used in the literature [49]. Among the optimization methods, mixed-integer linear programming (MILP) is the most frequently used one, followed by mixed-integer quadratic programming, non-linear programming and dynamic programming. The main metaheuristic methods are Monte Carlo simulations, genetic algorithms, agent-based models, particle swarm optimization, and game theory simulation.

In terms of energy community research, self-generation and prosumership are central aspects of any EC because, without generation technologies, an EC would collapse into just a simple aggregation of consumers. For this reason, many authors have focused their attention on the investigation of energy community technological configurations. Most studies are based on renewable energy generation technologies, even though traditional generation sources have often been included in EC technological portfolios. Storage technologies have instead mainly

Table 1

Legislative framework pertaining to the adoption of Collective Self-Consumption (CSC), Renewable Energy Communities (RES) and Citizen Energy Communities (CEC) in EU member states.

	AT	BE	BG	HR	CY	CZ	DK	EE	FI	FR	DE	EL	HU	IE	IT	LV	LT	LU	MT	NL	PL	PT	RO	SK	SI	ES	SE
CSC																											
RES																											
CEC																											

(white): none; (orange): proposal or draft; (yellow): general law framework; (blue): pilot projects; (green): national laws in place.

Table 2

Energy community support mechanisms currently in place in the EU member states and in the UK.

Country	Beneficiary of the supporting scheme	Incentivized energy quota	Period	Type of Incentive Mechanism	Premium tariff	Ref.
Belgium (BE)	Members (net billing)	a) Shared energy	¼ h.	a) Virtual net-metering	a) € 187/MWh (Retail price Eurostat 2020)	[40]
Germany (DE)	EC management	a) Shared energy b) Energy fed into the grid c) Network charges	1 h.	a) Virtual Self-consumption b) FIT c) Exemptions	a) € 23.7/MWh (<750 kW) € 37.9/MWh (<10 kW) € 121.1/MWh b) Market price c) € –80/MWh	[41]
Greece (EL)	Members (net billing)	a) Shared energy	1 h.	a) Virtual net-metering	a) € 129/MWh (Retail price without taxes -Eurostat S12020)	[42]
Italy (IT)	EC management	a) Shared energy b) Energy fed into the grid c) Network charges	1 h.	a) Virtual Self-consumption b) Net purchase and sales c) Rembourse d) FIP	a) € 110/MWh b) Market price c) € 9/MWh	[36]
Ireland (LT)	EC management	d) Energy generation	–	d) FIP	d) € 104/MWh (average RESS 1 Community Auction)	[43]
Portugal (PL)	Members (net billing)	a) Shared energy b) Energy fed into the grid c) Network charges	¼ h.	a) Virtual net-metering b) Bilateral contract c) Exemption	a) € 117/MWh (Retail price without taxes- Eurostat S12020) b) Selling price c) € 60, €80/MWh (CIEG Low Voltage)	[44]
Spain (ES)	Members (net billing)	a) Shared energy b) Energy fed into the grid (optional)	1 h.	a) Virtual net-metering b) Net purchase and sale	a) € 117/MWh (Retail price without taxes Eurostat S12020) b) Market price	[45]
The United Kingdom (UK)	EC management	e) Rural Community Energy Fund (RCEF) projects	–	e) Grants	e) k£ 40 (Stage 1) k£ 100 (Stage 2)	[46]

been investigated as a means of reducing the exchange of energy with a public grid, i.e., the amount of withdrawn energy and the generation surplus fed into the grid, or to offer flexibility rather than capacity service to the grid [50,51].

Energy communities have often been investigated in the literature as electricity-based energy systems, thus inheriting many aspects of smart grid research [12,25], even though multi-vector energy communities are the most likely configuration in real projects [34]. Many authors have shown increasing interest in the sector-coupling of energy communities [52], matching such different distribution grids as power, gas [53], heat [54], and hydrogen [55]. In this regard, such energy conversion technologies as heat pumps [56], electrolyzers and fuel cells [57], co-/tri-generators [58] are the main ones included in the technology mixes of ECs.

2.4. Sharing mechanism

P2P and virtual metering energy policies both enable different logics of energy communities and business model configurations [6,59]. Energy communities can engage in multiple energy services, thus generating additional profits, apart from selling self-generated energy. Okur et al. reviewed the main business models related to the participation of P2P EC on the energy market: trading flexibility on the day-ahead market, trading flexibility on the intra-day market, providing power reserves, balancing portfolio internally, or managing congestion. As reviewed in Section 2.2, a country-specific energy policy framework determines who receives the energy service revenues, the energy

communities or the EC members directly. In both cases, the energy community might establish two different energy trading levels: the external market, where the EC energy services are sold [30], or an internal market, which is used to redistribute the generated economic benefits among the EC members according to a specific set of rules, herein called *sharing mechanism* (SM).

The sharing mechanisms in the EU directives are intentionally not disciplined, thereby leaving member states or energy community members the opportunity to define how to use and redistribute the EC profits according to an internally adopted scheme. Some countries, such as Belgium, France, and Spain, have proposed a national sharing mechanism framework whereby the shared energy is directly distributed to the EC members by the DSO through net billing. Nevertheless, these sharing mechanism cases have not been considered in this work since they can lead to an internal third-level energy trading among EC members to implement a custom-built compensation system that is not considered in the DSO calculation. For example, to balance the compensation among all members even though the direct remuneration is received only by those members to whom the RES generation plant is connected. For a similar reason, P2P schemes are not considered in this work either.

In general, the structure of the sharing mechanisms mainly depends on the EC business model, and each member contributes to achieving the EC goals and objectives. For example, an energy community that adopt a business model to trade flexibility on the intra-day market might distribute the EC profits among the members according to their contribution to shifting loads. Conversely, an energy community that invests

in a new power plant to receive a FiT or FiP tariff might distribute profits proportionally according to the members' investment contributions.

In literature, sharing mechanisms have only been investigated in the context of Energy Communities that are based on a P2P policy framework [60–62]. The two main approaches are based on the Game Theory [63] or MILP [64]. However, little research has been conducted on EC based on virtual net-metering [65], whose techno-economic benefits have only been investigated collectively, rather than individually, for each member [58,66]. Considering that most of the energy communities that have emerged in Europe are based on virtual metering, the lack of a deep and detailed investigation on the sharing mechanism in this energy policy framework results in a significant scientific literature gap. Moreover, a general understanding of the mechanisms adopted to fairly distribute the benefits among the members is a prerequisite to engage citizens in these projects.

Moncecchi et al. have recently used a Shapely method in a game theory simulation to determine the added value of members of an energy community. This method is very effective in determining the user-added value and the EC marginal benefits. However, the use of the mathematical formulation of the Shapely method is the main drawback of this method. The computational burden already becomes challenging for energy communities with some hundreds of members [67]. Moreover, the algorithm is not straightforward for final users who want to understand how to modify the consumption behavior in order to increase the shared energy. One of the aims of this work is to systematically investigate the sharing mechanism of an energy community in a virtual net-metering energy policy framework. For the first time, a straightforward methodology has been used to define how the economic benefits (profits) are shared among EC members based on the service they provide to the EC. In fact, EC members can provide a financial service by owning the RES generation assets, or an operational service by engaging in the self-consumption of RES electricity, or even both. Three different sharing mechanism principles are therefore proposed, and their impact on different kinds of residential users is assessed.

3. Method

3.1. Simulation framework

Energy community modeling involves simulating the exchange of dynamic energy between all its nodes: demand nodes (e.g., EC members), generation nodes (e.g., renewable power plants), generation and demand nodes (e.g., prosumers) and storage nodes. Time-dense data with (at least) an hourly resolution is generally needed. Such data could be historically measured data or synthetic data. Hourly generation profiles from PV installations can be accurately modeled with such tools as PVGIS [68]. Measured real data, or synthetic data (see, for instance, the work of Bottaccioli et al. [69]) can be used for the load profiles of residential users. Such high-frequency datasets are required to evaluate the economic value of all the energy exchanges. They are needed, for example, to account for the EC shared energy awarded for the virtual net-metering tariff considered in this work.

The simulation framework used in this work is based on a modular Python architecture, as described by Minuto et al. [70]. Electricity load demand (D-El), photovoltaic demand (T-PV), and energy balance (T-EB) modules have been employed in this work to simulate the EC energy flow with an hourly resolution. The D-El module generates a synthetic electric load demand for each residential user. The T-PV module implements the PV-GIS tool to simulate the generation profile of a rooftop photovoltaic system. The T-EB module calculates the energy balances for all the EC nodes for a specific grid aggregation level, such as node, building, substation feeder, MV/LV substation, etc. (see Fig. 2). All the scenarios described hereafter have been evaluated by simulating a complete year at an hourly resolution, i.e., 8760 h.

The total amount of the energy generated (E_g) at time step (t) is the sum of the energy generated (G) by the j -th generation plant for all the nodes (n_i) for aggregation level (α):

$$E_g^{\alpha}(t)[kWh] = \sum_{n_i^{\alpha}} \sum_j G_{ij}^{\alpha} \text{ where } n_0^{\alpha}, n_1^{\alpha} \dots n_i^{\alpha} \in \alpha \quad (1)$$

similarly, the energy demand (E_d) is defined as the sum of all the

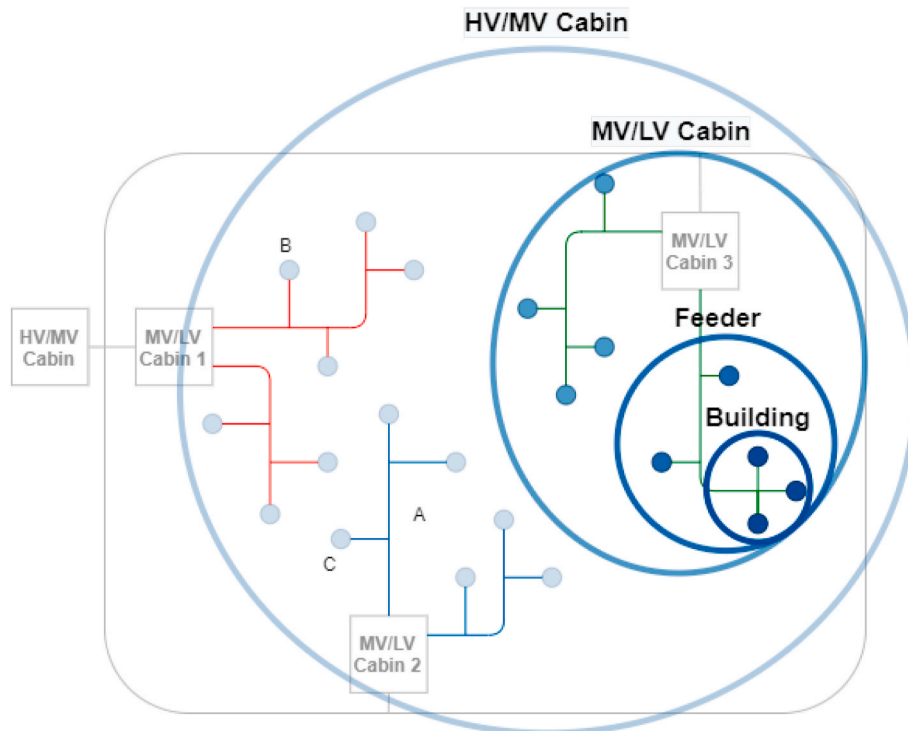


Fig. 2. Schematic representation of a network grid and different aggregation levels of the nodes (Blue circles).

energy demands (D_i) of the i -th node:

$$E_d^\alpha(t)[kWh] = \sum_{i \in \alpha} D_i^\alpha \quad (2)$$

By combining Eq. (1) and Eq. (2), it is possible to calculate the energy withdrawn (E_w) and fed into the grid (E_f), and the Shared Energy (SE), as:

$$E_w^\alpha(t) = \text{abs}\left(\left[E_g^\alpha(t) - E_d^\alpha(t)\right]\right)_{<0} \quad (3)$$

$$E_f^\alpha(t) = \left[E_g^\alpha(t) - E_d^\alpha(t)\right]_{>0} \quad (4)$$

$$SE^\alpha(t)[kWh] = \min\left(E_d^\alpha(t); E_g^\alpha(t)\right) \quad (5)$$

The shared energy is the minimum amount (on an hourly basis) between the energy self-generated by the EC that is exported to the local distribution grid and the overall and collective electricity withdrawn from the grid by community members. Whenever a community member acts as a prosumer, the self-produced energy is first self-consumed by themselves. The remaining quota, if any, is exported to the grid and is eligible to become shared energy. We thus call the amount of energy that is not exchanged with the grid by the single prosumer belonging to the EC “physical self-consumption”. Shared energy is instead virtual collective self-consumed energy.

Different simulated energy flows are provided in Fig. 3 for a week of the considered year to understand the accounting of the various energy flows relevant to the EC. Whenever there is net export to the grid, the stack plot shows the overall generated energy (which the sum of green and yellow areas); at times of net import from the grid, the graph shows the overall energy demand (the sum of green and pink areas).

3.2. Business model and ownership schemes

The business model of the energy community modeled here is based on a maximization of the incentives. The incentives considered in this work are granted to an energy community manager for the *shared energy* (SE) distributed among the EC members in a virtual net-metering scheme. Therefore, the EC members do not receive the incentives

directly, and two-level trading is required to redistribute the generated revenues among the members. The energy community manager obtains revenues, from the first level trading, that are proportional to the SE :

$$Revenue_{EC}[\text{€}] = SE[kWh] * Incentive[\text{€} / kWh] \quad (6)$$

$$SE = \sum_t SE^\alpha(t) \quad (7)$$

where α (the EC aggregation level) and t (the time window, e.g., 1 h) are set in the energy policy.

Within the energy community, the second level of trading defines how the EC net profits (P_{EC}) are distributed among the members, using a sharing mechanism algorithm. P_{EC} is determined according to the specific financial plan of the energy community and can be formulated in a general form as:

$$P_{EC}[\text{€}] = Revenue_{EC} - Costs_{EC} - Accumulation fund_{EC} \quad (8)$$

A practical way of distributing the EC net profit (P_{EC}) is through resort to an internal Token currency (TC):

$$P_{EC}[\text{€}] = TC[TC] * A \left[\frac{\text{€}}{TC} \right] \quad (9)$$

where the conversion factor, A , might simply be set equal to one. The TC gained by the i -th member is proportional to the shared energy (SE_i) this member contributed, weighted by an internal distribution factor, f_i :

$$TC_i = SE_i[kWh] * f_i \left[\frac{TC}{kWh} \right] \quad (10)$$

$$\text{with } \sum_{i=1}^{n \text{ member}} f_i = 1 \quad (11)$$

where distribution factor f_i , can eventually be used to assign a different economic value to the SE of specific users (e.g., vulnerable users). The SE_i of each member is calculated by specific sharing mechanism algorithms, which may have a different rational. Three possible algorithms are introduced in the following section.

It is worth mentioning here that the Token currency is just an

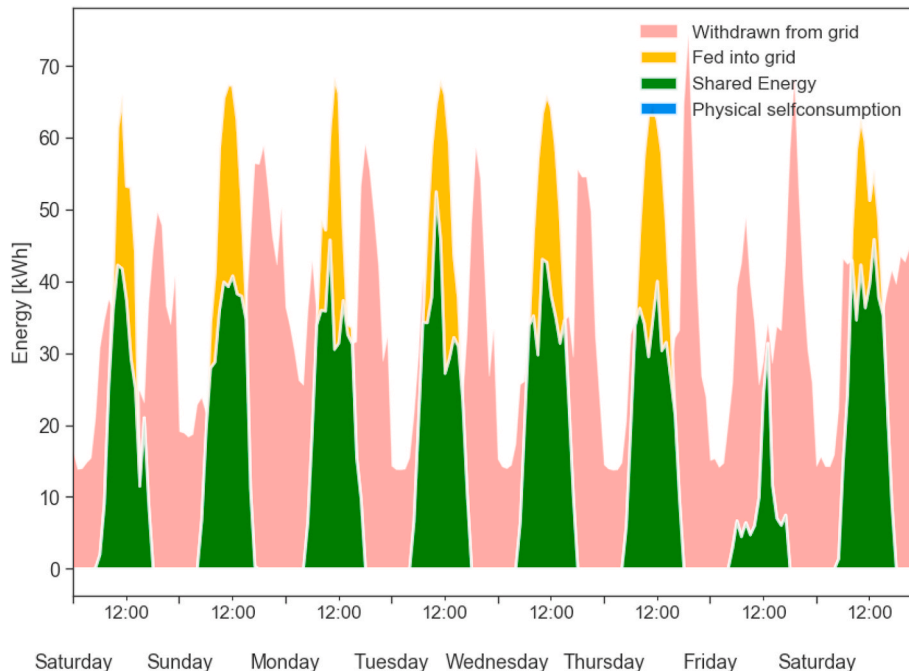


Fig. 3. Energy flows for an energy community (a week sample). Physical self-consumption is zero when the generation node does not have any load demand.

internal currency that is used by an energy community to remunerate its members. In a real-world setting, the Token currency could be replaced by either an actual currency (e.g., Euros) or remain an internal accounting system to remunerate members that could then trade the token to access other (energy or non-energy) services within the energy community.

Another important aspect of the EC business model is the ownership of the RES power generation plant and the related financing mechanism. Table 3 shows a possible EC ownership classification with respect to the type of ownership-financing (OF) mechanism.

In Type-B and Type-C ownership schemes, the energy community owns the generation plants, and the funds come from the EC members, who either share the initial investment equally (Type-B) or based on an individual quota (Type-C). These schemes imply that when an EC member decides to leave the community, the generation plant remains available to the other members of the energy community. On the other hand, the members directly own the generation assets in the Type-E and Type-F ownership schemes. Therefore, an EC member who decides to leave the community either sells their share to the other EC members or is open to potential disputes on co-ownership rights and duties as a result of continuing to own their generation plant share. The Type-D ownership scheme refers to the case of an energy community that exploits one or more of the generation plants individually owned by some EC members (prosumers). In such a case, whenever an EC prosumer member leaves the community, the EC either loses the availability of the power generation or the former EC member becomes a third-party energy producer for the EC, i.e., the OF scheme of the EC becomes a Type-G.

In the Type-A and Type-G schemes, the energy community uses the power plants without paying for the initial capital investment. In the former case, the EC acquires the ownership of the generation plant by utilizing some financing mechanism (i.e., a bank loan, a national fund loan, grant, etc.), while in the Type-G scheme, a third party is the owner of the asset, i.e., an energy service company or an energy utility. Both ownership schemes imply extra operational costs, as can be seen in Eq. (8), and the net profit, P_{EC} , is therefore reduced to either pay back the loan (Type-A) or lease the assets (Type-G).

3.3. Sharing mechanisms

The aim of the sharing mechanism is to fairly redistribute the Token currency among the EC members according to the services they provide to the community. In the case of the virtual net-metering energy community investigated here, these mechanisms are:

- *self-consumption services*, which are the contributions made to maximize the shared energy in order to consume the energy in the same time window as the renewable energy is generated (*OPEX-type*);
- *financing services*, which are investments for the installation of new renewable generation plants to supply the EC (*CAPEX-type*).

Table 3
EC ownership classification with respect to the type of ownership-financing mechanism.

Ownership \ Financing mechanism	Investment fund	EC Members (equal shares)	EC Members (individual quota)	Leasing
Energy Community	Type-A	Type-B	Type-C	*
EC Members (individual-ownership)	*	–	Type-D	*
EC members (co-ownership)	*	Type-E	Type-F	*
Third-party	*	*	*	Type-G

* beyond the scope of this paper

This work investigates the impact of the adopted sharing mechanism on an energy community that has different types of ownership and financing schemes for the energy generation assets. Three typologies of sharing mechanism algorithms that distribute the Token currency among the members proportionally to the services provided by each member to the EC are presented hereafter: *financing service* (SM1), *self-consumption service* (SM2), and both *financing and self-consumption services* (SM3).

3.3.1. Sharing mechanism 1 (SM1)

The SM1 sharing mechanism distributes the Token currency among all the members according to each member's ownership share of the generation power plant, regardless of their actual contribution to the shared energy of the EC. Thus, SM1 does not consider the members' consumption profiles or their efforts to maximize the amount of shared energy for the community. The i -th member's SE is calculated as:

$$SE_i = SE * Ow_i \text{ where } Ow = \sum_{i=1}^N Ow_i = 1 \quad (12)$$

The i -th member's shared energy (SE_i) is calculated by multiplying the SE by the user's Ow_i factor, which represents the ownership share of a RES power plants by the i -th EC member. In general, the Ow_i factor can change from user to user, or between different categories of members, e.g., consumer vs owners. In a simplified financing scheme, where all the EC members ($\#N$ members) have the same ownership share (e.g., Type-A OF), Ow_i is equal for all the members, and is equal to $1/N$. Therefore, the shared energy associated with each user is equal for all the members and SE/N values.

3.3.2. Sharing mechanism 2 (SM2)

The SM2 sharing mechanism distributes the Token currency among all the members proportionally to their actual contribution to self-consume the energy generated by the EC power plants. Since the shared energy is virtually self-consumed, there are no physical bases to calculate the energy exchange among the users; therefore, a resource allocation algorithm is required to calculate each user's shared energy (SE_i).

SM2 distributes the shared energy $SE(t)$ progressively among all the members for each billing period t (e.g., 1 h) up to their maximum energy demand. However, SE_i does not simply correspond to the user's energy demand. In fact, by ordering the users ascendingly according to their energy demands at a given time (t), the EC shared energy results to be the green area depicted in Fig. 4, while the gray columns are the energy demand that is not covered by the RES and withdrawn from the grid. The implementation algorithm is presented in Appendix B.

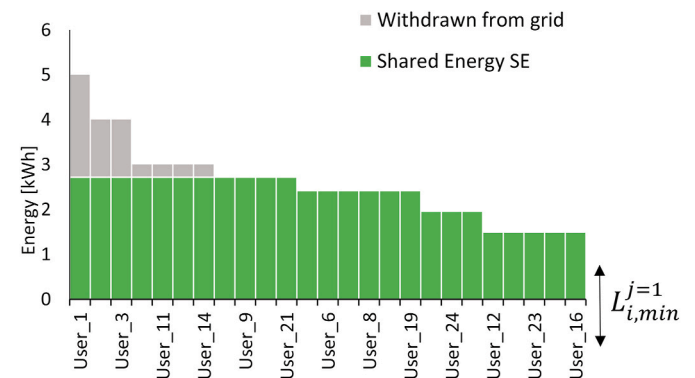


Fig. 4. SM2 shared energy allocation calculated for hour t for all the EC members.

3.4. Sharing mechanism 3 (SM3)

The SM3 sharing mechanism distributes the Token currency considering both services that an EC member could provide, i.e., the financing service and self-consumption service.

In this sharing mechanism, the energy self-generated by a specific power plant is first self-consumed (virtually) by its owners. Then, if the energy generation exceeds the owners' energy demand, it is shared among all the other EC members. Therefore, the i -th EC member's energy share, SE_i is constituted by two terms:

$$SE_i = SE_i^{SM1} + SE_i^{SM2} \quad (13)$$

where, the first term calculates the shared energy allocated when using SM1 ($SE_{i,BS1}$) because of the ownership rights, and the second term allocates the SE to SM2 ($SE_{i,BS2}$), because of the provided self-consumption service.

This sharing scheme implies that the EC members are treated differently according to whether they have contributed to financing the power plants or are just consumer members of the EC. For instance, a condominium might decide to install a PV system. However, not all the members might decide to contribute to financing it, even though they agree to participate in the energy community. Therefore, the energy generated by the building PV system is first reserved for its owners (SM1), and the generated surplus is then shared among the remaining EC consumer members through the SM2 algorithm. It is important to underline that in the first SM1 allocation, which assumes equal ownership, $O_{wi} = 1/N$, each of the owners has a self-consumption priority on a fixed quantity of SE_i that is equal to SE/N . For this reason, at a certain time step, if the load demand of the owners is lower than SE_i , the difference is distributed among all the EC members in the second allocation step using the SM2 algorithm. Instead, if the owners' load demand is higher than SE_i , the exceeding load does not have self-consumption priority and will be evaluated for a further contribution in the second allocation step.

3.5. Case study definition and EC scenario design

This section defines the case study and describes the design of EC scenarios. The used approach is that of simulating the yearly energy flow of an energy community of residential users who are powered by a photovoltaic plant connected to the main grid under a virtual net-metering policy framework. By evaluating the energy withdrawn by each member and the electricity fed into the PV plant grid, it is possible to calculate the shared energy of the EC for an hourly resolution. Then, applying the different algorithms proposed in Section 3.3, it is possible to investigate how the EC revenues are spread among the different types of users, depending on the adopted sharing mechanism and the applied ownership-financing scheme.

A good match between the energy demand and renewable power generation profiles of the EC members is generally the key to making Energy communities successful. The first step is to design an energy community in which the RES generation can be entirely self-consumed. Thus, in this work, the energy generation-to-energy demand ratio is 1/3. However, because of the non-programmability of RES, the coincidence between energy demand and generation, in the absence of storage systems, depends on the EC members' energy consumption profiles. For this reason, the EC has been designed to be populated by members with different annual demands and load profiles in order to investigate the members' self-consumption services. However, their contribution to financing the power plant (i.e., the financing service) has to be considered. In order to investigate what type of users benefit from the different SM algorithms, different ownership EC scenarios have been generated.

The physical domain of the case study is an urban neighborhood. It comprises 100 residential users and one photovoltaic power plant, all of which are connected to the same transformation cabin (same α aggregation level as in Eq. (1)). The members are distributed among

residential buildings; one of the buildings hosts a PV system with 100 kWp of rated power, mounted at a roof-top pitch of 20°, and the tilt directed as follows: 50 kWp south, 25 kWp east, and 25 kWp west. The users' electric load profiles are generated using the Italian *Time-of-Use* (ToU) tariffs; the PV system is georeferenced for Italy. The users' total annual energy demand is shown in Figure A-1, and the PV settings are reported in Table 4.

The EC members are divided into four *user type* (UT) groups, on the basis of their load demand characteristics. Each group comprises 25 households, categorized according to two main key performance indicators (see Fig. 5-a): the *Pearson correlation coefficient* (PCC), which measures the temporal match between the users' demand profile and the PV generation profile over the whole year, and the production-demand ratio, which is defined as the ratio between the PV annual energy generation per member and the annual demand of each member, and it discriminates between high demand and low demand users. The annual energy demand of the energy community, for an hourly resolution, is shown in Figure A-2 as a stacked area plot where each shade of color represents the load demand of an EC member. The distribution of the residential EC members in the four User Type (UT) groups is shown in Fig. 5-a. Quadrant Q1 represents the EC members who have a low energy demand and low level of synchronism between their load profile and PV generation. Members belonging to the Q3 quadrant have a low load demand, but high load-production synchronism. The Q2 and Q4 users have a high load demand, with the Q4 users exhibiting a high PCC, unlike the Q2 users. Fig. 5-b shows the load profile stacked plot of the EC members for each UT group. The PV generation profile is shown as a dashed black line with a 25% scaling factor for comparison purposes. Fig. 5-b, which refers to the yearly time-series on a weekday in August, offers an insight into the synchronism between the PV generation and each EC member group. The Q1 and Q2 UT groups exhibit a load demand peak during the night hours. For this reason, these UT groups have a negative PCC, while the Q2 and Q4 UT groups show a substantial energy demand during daylight hours, thus leading to a positive PCC.

Having set the energy community generation and demand behaviors, six EC scenarios were defined to investigate the three proposed sharing mechanisms (described in Section 3.3) for different PV member ownership schemes. The ownership of the PV plant is distributed equally among all the members in Scenarios S1 and S2, which correspond to SM1 and SM2, respectively. The goal is to understand the impact of the different SMs, without considering the ownership aspects. The PV ownership in EC scenarios S1 and S2 is shared equally among all the EC members, i.e., O_{wi} is equal to 1/100 for each member. Scenarios S3 introduces the ownership aspect onto the scene. Therefore, we explored sub-cases in which the ownership of the plant was just attributed to one user type, while the other EC members (without co-ownership) offered their energy load to the calculation of the shared energy. Here, the aim was to understand how ownership (which entails a priority on the physical self-consuming of the generated electricity before sharing it among community members) could be beneficial for certain UTs, and to what extent. Since ownership in the S3-Q# scenarios belongs to one UT group at a time, O_{wi} is equal to 1/25 for the owners and equal to zero for the non-owners. In all the scenarios, the members' shared energy is converted into Token currency using Eq. (10), where the weighting factor, f_i , is chosen to be equal for each member ($f_i = 1/100$). Table 5

Table 4

Details of the photovoltaic system considered for the Case Study.

	Roof Pitch-1	Roof Pitch-2	Roof Pitch-3
PV Nominal Power	50 kWp	25 kWp	25 kWp
Equivalent hours	1305 kWh	1088 kWh	1094 kWh
PV power density	200 W/m ²		
PV azimuth	0° (south)	−90° (east)	90° (west)
PV tilt angle	20°		
Rooftop coverage factor	70%		
Location (Lat, Lon)	(45.067367, 7.656817)		

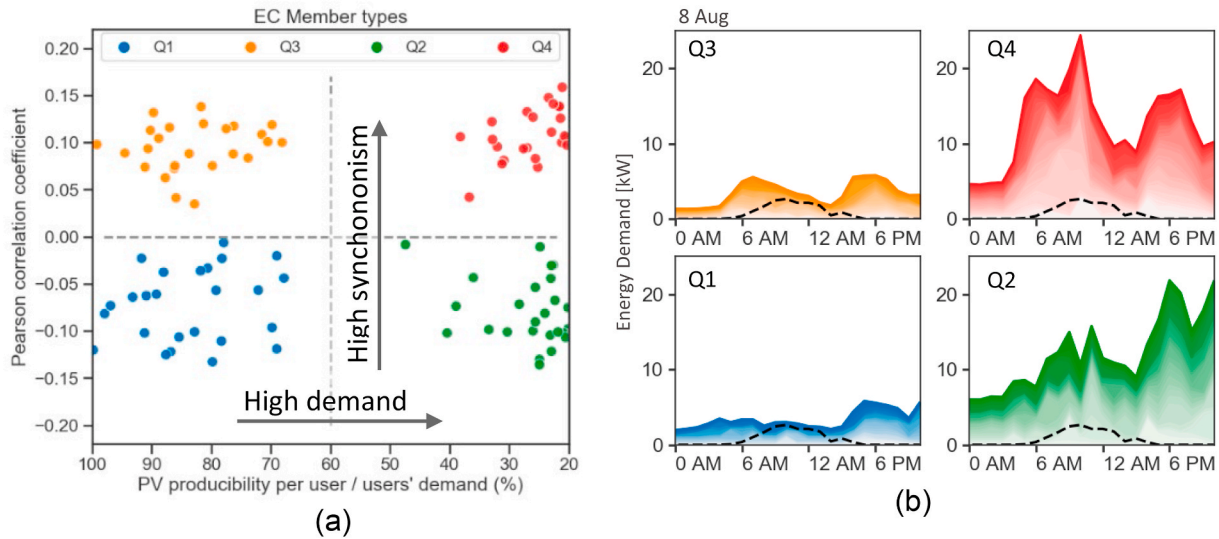


Fig. 5. (a) Classification of EC members into four groups; (b) Stack plot of the 25 load profiles, for each EC group (each shade of color is a different member's load profile), for a weekday in August, compared with the PV load profile scaled by 25%.

Table 5

Overview of the simulated scenarios. A solid black circle indicates that the EC group members are the co-owners of a PV plant (ownership is equally distributed among all the members for S1 and S2).

Scenario	Sharing mechanism	PV co-ownership of the UT group members			
		Q1	Q2	Q3	Q4
S1	SM1	•	•	•	•
S2	SM2	•	•	•	•
S3-Q1	SM3	•			
S3-Q2	SM3		•		
S3-Q3	SM3			•	
S3-Q4	SM3				•

summarizes the scenarios discussed above.

4. Results and discussion

The results of the energy flow simulation within an energy community are reported in Table 6. A community could self-consume all the RES generated electricity for a PV plant that generates about one-third of the EC energy demand. Nevertheless, only 77% of the electricity was found to be collectively self-consumed, as 27 MWh/a of energy was sold to the grid. The actual shared energy is 90 MWh/a, thus a value of collective self-sufficiency (SE/E_d %) of 29% is achieved. The PV power plant grid node was set to not have any electricity demand (zero physical self-consumption). Thus, the energy withdrawn from the grid, E_w , is equal to the energy demand, E_d , and the energy fed into the grid, E_f , is equal to the generated energy, E_g .

The SM1 sharing mechanism (financing service) was investigated by simulating the S1 scenario. The energy community designed for this

scenario shares the PV plant ownership rights equally among all the EC members and UT groups. The results in Fig. 6-a show the Token currency gained by each EC member over the whole year. In this graph, the EC members are aggregated in subplots on the basis of their user type (UT) group. It can be noted that all the members of all the UT groups receive the same amount of Token currency, i.e., they have the same benefits, regardless of their demand profile or their annual energy demand. On one hand, this allocation might sound fair and equalitarian, but the main drawback of the SM1 sharing mechanism is the absence of feedback on the member's demand behavior (i.e., demand profile). In other words, this remuneration scheme does not promote a virtuous energy consumption. In fact, a member who decides to modify their demand curve to increase the shared energy of the EC will not receive any extra rewards as a result of their behavior, as any extra profit gained by the EC is spread equally among all the members.

The Token allocation mechanism, which is based on the SM1 algorithm, could be relevant for an energy community in which the EC members all have the same ownership share of the generation assets. Bearing in mind the EC classification introduced in Section 3.2 (Table 3), the Type-B, and Type-E energy community configurations are those where each EC member has financially contributed equally to the realization of the power plant. Alternatively, as for the Type-A and Type-G ownership schemes, a third party finances the generation assets with grants (from a national institution), leasing services (from private service companies), or loans (from banks). Instead, in energy communities in which the PV plant ownership is not distributed equally among the members (e.g., for the Type-C and Type-F schemes), the members who financed the generation assets receive the same economic benefits as others who did not contribute to it in any way. At a first glance, this result might appear unjust. However, the fairness of the allocation principle depends to a great extent on who the involved members are, the energy community goals, the mission, and the ethics. For example, imagine an EC where the *consumer* members are fragile families (with a low or limited energy demand because they cannot afford to pay the energy bill) and the goal of the EC is to fight against energy poverty. In such a case, the SM1 sharing mechanism will support those families that are in an energy poverty condition, because the economic benefits are not proportional to their limited energy demand, which might not be synchronized with the generation plan. Thus, such a mechanism could result to be ethically fair.

Table 6

–Simulation results: EC energy flow.

	Annual Energy (MWh/a)
Electricity generation (E_g)	117
EC electricity demand (E_d)	309
Physical self-consumption	0
Shared Energy (SE)	90
collective Self-Consumption	77%
collective Self-Sufficiency	29%

TC allocation principle: the members' financing service (SM1).

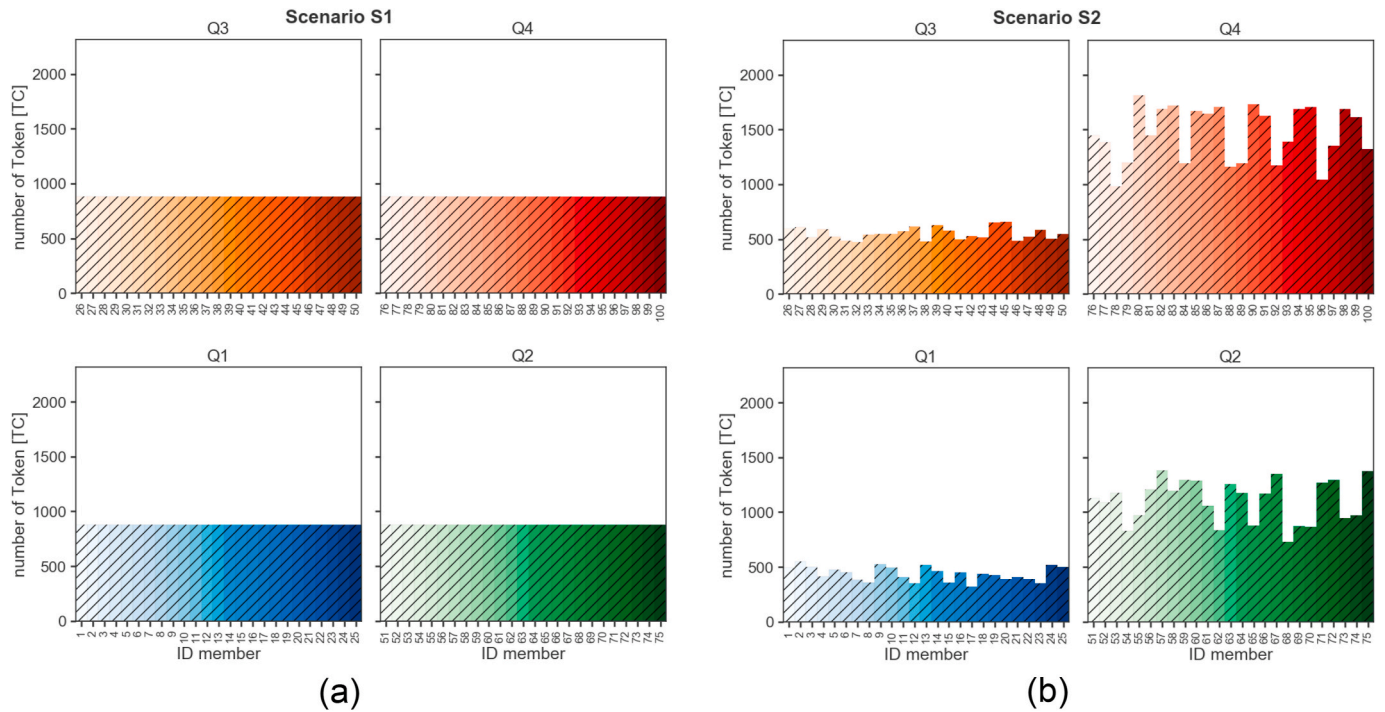


Fig. 6. Allocation of a Token currency to members of the different UT groups, according to: (a) the SM1 sharing mechanism, i.e. scenario S1; (b) the SM2 sharing mechanism, i.e. scenario S2. The hashed columns indicate the quota of Tokens allocated because of ownership priority. Each shade of color indicates a distinct member within the same UT group.

4.1. TC allocation principle: the members' self-consumption service (SM2)

The S2 EC scenario investigates the self-consumption service sharing mechanism (SM2). All the EC members are considered to be generation asset owners. The TC is distributed equally and progressively among those who contribute to maximizing the shared energy (see Fig. 6-b). The plot aggregates the EC members into subplots on the basis of their user type group. Fig. 6-b shows that each member gains a different amount of TC, depending on their energy demand behavior. The Q2 and Q4 members obtain more TC than the Q1 and Q3 UT group members, because of their higher total energy demand (see Fig. 5-a). Moreover, the Q4 user type group receives, on average, slightly more TC than the Q2 group because of a high temporal generation-demand synchronism (positive Pearson Correlation Coefficient in Fig. 5-a). Similarly, the Q3 members obtain more TC than the Q1 ones, even though this is less evident in the graph. These results are highlighted in Fig. 7, where the difference in TC received in the S2 scenario is compared with the TC received in the S1 scenario.

The SM2 sharing mechanism mostly penalizes the low energy demand UT groups (Q1 and Q3), and, in general, those members who have an energy demand that is temporally not synchronized with the availability of the power generation. This result is not affected by a specific RES generation asset choice, because the Pearson Correlation Coefficient relates the temporal coherence of the energy demand to the production, regardless of the specific RES generation pattern. What changes when other RES assets are considered, e.g., wind turbines, is that the generation pattern might promote the need for self-consumption in different hours or seasons. The energy demand of those members who perceive more TC peaks in the central hours of the day, when the EC is supplied by solar energy, is considered in the current EC scenario.

The SM2 sharing mechanism, which only accounts for the Energy Shared by the actual members, does not discriminate between owner and non-owner members, and therefore just rewards the synchronism between the energy demand and generation profile. Thus, the higher the

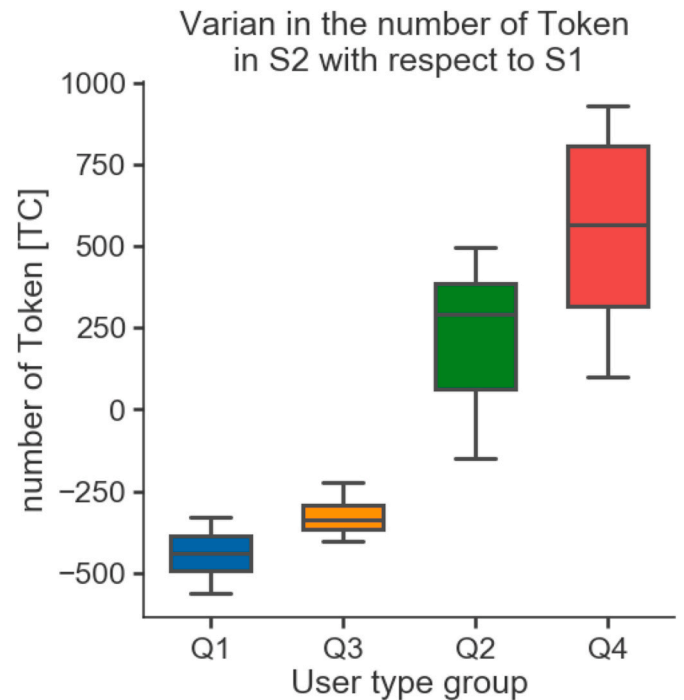


Fig. 7. Net Token difference among the UT group members in the S2 scenario with respect to the S1 scenario.

member's energy demand and PCC are, the higher the TC they receive. It should be noted that a Q1 or a Q3 member receives as much as -80% TC than a Q2 or a Q4 member. The SM2 distribution scheme might be considered discriminatory or unfair for energy communities in which all the members contributed to financing the generation plant (Type-B or Type-E ownership schemes). Instead, this allocation algorithm might be

suitable for Energy communities that have a Type-A financing scheme (loan or funds) or Type-G ownership (a third party that provides the generation service) because they obtain economic benefits without investing personally in the generation assets. Moreover, it is possible to use the SM2 mechanism for ECs based on Type-C schemes, which are characterized by an individual share of asset ownership for each EC member, to make the annual financing quota of the members proportional to its Token currency.

4.2. TC allocation principle: the members' financing and self-consumption service (SM3)

The last presented sharing mechanism is SM3, which was designed to distribute the TC among the members according to their financing service and self-consumption service. This allocation algorithm gives self-consumption priority to those EC members who are owners of the generation assets. The SM3 allocation algorithm is the same as the SM2 one, but is composed of two steps. In the first round, the shared energy is attributed primarily to owners-members. If any available RES energy is

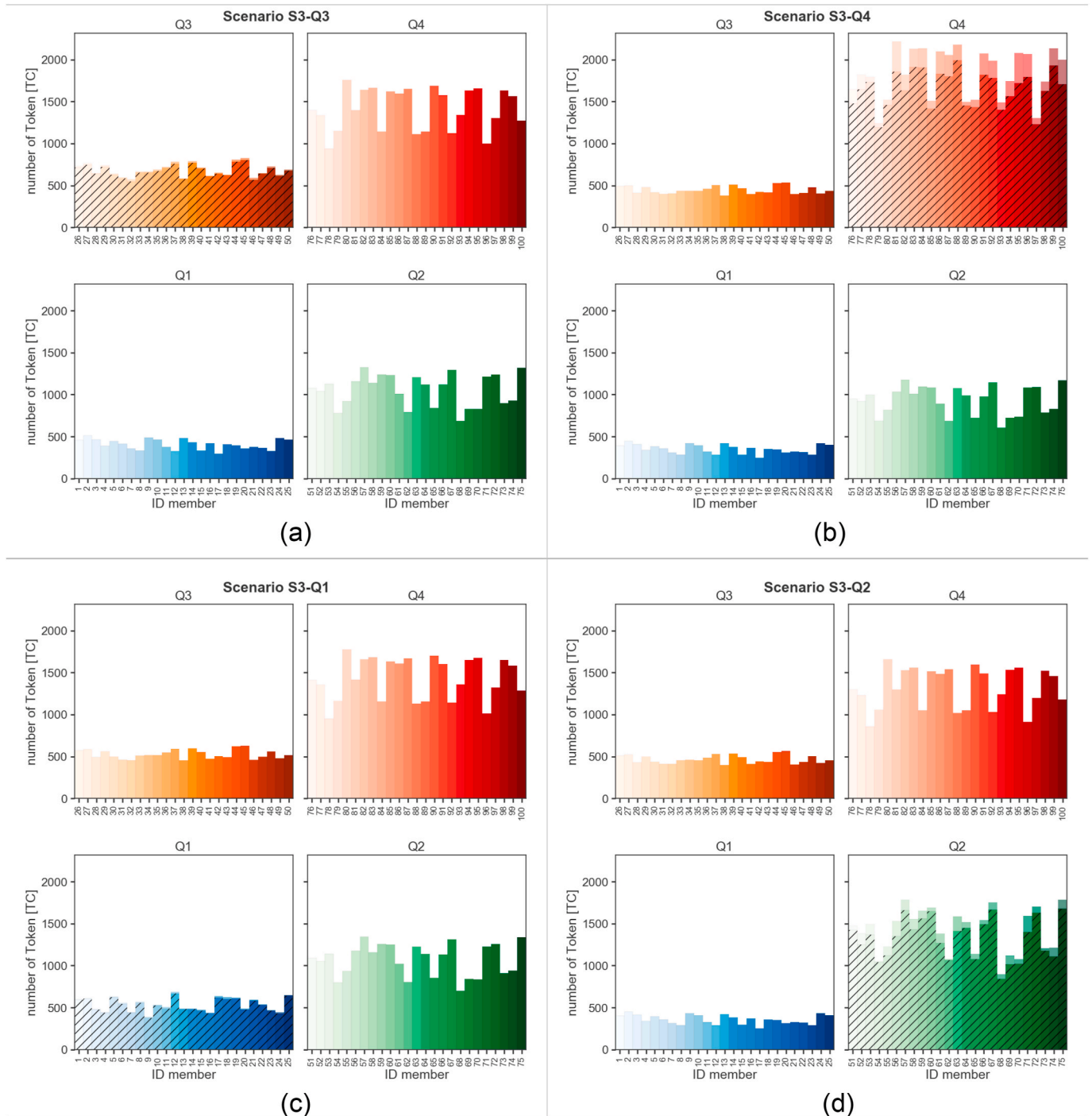


Fig. 8. Allocation of Token currency to members for the different UT groups, according to the SM3 sharing mechanism: (a) scenario S3-Q3; (b) scenario S3-Q4; (c) scenario S3-Q1 (d) scenario S3-Q2. The hashed columns indicate the quota of Tokens allocated according to an ownership priority. Each shade of color indicates a distinct member within the same UT group.

left over from the first round, it is shared equally among all the EC members in the second step. With this allocation mechanism, an owner member can obtain Token currency for both financing and self-consumption services. Whenever a member's energy consumption exceeds SE_i in the first round, they can obtain more TC in the second round. On the other hand, pure consumer EC members can only obtain TC in the second allocation round.

In order to investigate how the SM3 sharing mechanism affects the different UT groups in the different PV ownership schemes, four EC scenarios were investigated: S3-Q1, S3-Q2, S3-Q3, and S3-Q4. In each scenario, the PV ownership was assigned to the members of one UT group at the time. The results of these scenarios are compared in Fig. 8.

The hashed bar in Fig. 8 indicates the TC received by each member for the financing service, while the open bar indicates the TC received for the self-consumption service. It can be noted that only the user type group members who own the plant obtain TC for the financing service. In the S3-Q1 and S3-Q3 scenario, the owner groups, i.e., Q1 and Q3, are the groups that gain the most TC for the financing service (hashed bar). This result is due to the low energy demand of such members. On the other hand, in the S3-Q2 and S3-Q4 scenarios, the owner-members (Q2 and Q3) also receive TC for the self-consumption service calculated in the second allocation round (open bar). In fact, the Q2 and Q4 UT groups have such a high load demand that it saturates their available SE_i in the first allocation round, and they then obtain more TC in the second allocation round.

Fig. 9 helps identify what members benefit from the ownership of the generation assets under the SM3 sharing mechanism; the box plot compares the TC of the different UT group members in the S3-Q1, S3-Q2, S3-Q3 and S3-Q4 scenarios with respect to the S2 one.

Looking at the S3-Q1 scenario in Fig. 9, it appears that the TC of the Q1 UT group members that hold asset ownership on average increases by 100 units, while the other groups lose about 25–30 Tokens each. A similar result is registered in the S3-Q3 scenario, with the Q3 UT group as owner-members. On the contrary, the ownership of the Q2 UT group members leads to an increase in their Token currency, from 150 to 400 units, with an average increase of about 300 units, mainly at the expenses of the Q4 group, while in the S3-Q4 scenario, the increase in TC is even greater, that is, from 250 to 470 units, mainly at the expenses of the Q2 group. The strong competition between the two high-demand UT groups that gain TC at the expense of other parties in the scenario where they own the generation assets should be noted.

These results clearly show that the Q2 and Q4 UT groups have a greater advantage from the self-consumption priority, as the TC increases by up to 30% with respect to the S2 scenario. However, the Q4 group scores more TC than the Q2 UT group, because of its positive

Pearson Correlation Coefficient. Instead, the synchronism coefficient does not affect the economic performance of the low energy demand groups (Q1 and Q3) so much for the SM3 sharing mechanism, and loses 10% of TC, compared to the S2 scenario.

4.3. Who benefits from RES ownership in an EC?

Table 7 summarizes the results of all the EC scenarios and reports the average TC gained by each UT group, according to the investigated SM allocation schemes. SM1 results to be the best beneficiary sharing mechanism for low-demand UT groups, such as the Q1 and Q3 ones. On the other hand, Q2 and Q4 members benefit from the SM2 and SM3 algorithms, and achieve the highest TC value for the S3-Q2 and S3-Q4 scenarios, respectively, where they have the generation plant ownership rights.

In practice, the energy communities in which the members are not asked to pay for generation assets, i.e., Type-A and Type-G, can refer to Table 7 results to orient their choice of what sharing mechanism to adopt internally. When making their choice, the EC should consider the composition of the specific EC user type, the EC objectives and its mission. On the other hand, these results do not clarify the benefits the members can achieve for all the other energy community financing schemes, mainly where members contribute to financing the generation asset. In order to understand which UT group would benefit the most from a specific sharing mechanism and in which ownership-financing scheme, it would be necessary to relate the value of TC in Table 7 to the capital cost expenditure (CAPEX) of the initial generation assets. Thus, it would be possible to determine the net present value (NPV) of the investment at the end of the incentive or at the end of the lifetime of the generation asset.

Estimating the economic value of the Token currency and the CAPEX in local currency is not a trivial matter, as they depend on the specific country context in terms of incentives, electric price, market rule, and the CAPEX of the generation assets. To overcome this issue, and to obtain an insight into how the different ownership schemes affect the owners' net present value (NPV), it is possible, for example, to assume that the annual energy community net profit, P_{EC} , and the CAPEX of the initial assets lead to a payback time (PBT) of the investment of 10 years. Considering a total annual TC of 3520 gained yearly by the EC, an interest rate of 4%, and a PBT of 10 years, the equivalent cost of asset generation should be a Token currency of 28,850. Therefore, from Table 8, it is possible to evaluate the Token currency of the average NPV for the 25th year (equal to the PV lifetime) of a member belonging to a specific UT group in the different EC scenarios and OF schemes.

Table 8 shows the results of energy communities in which the EC

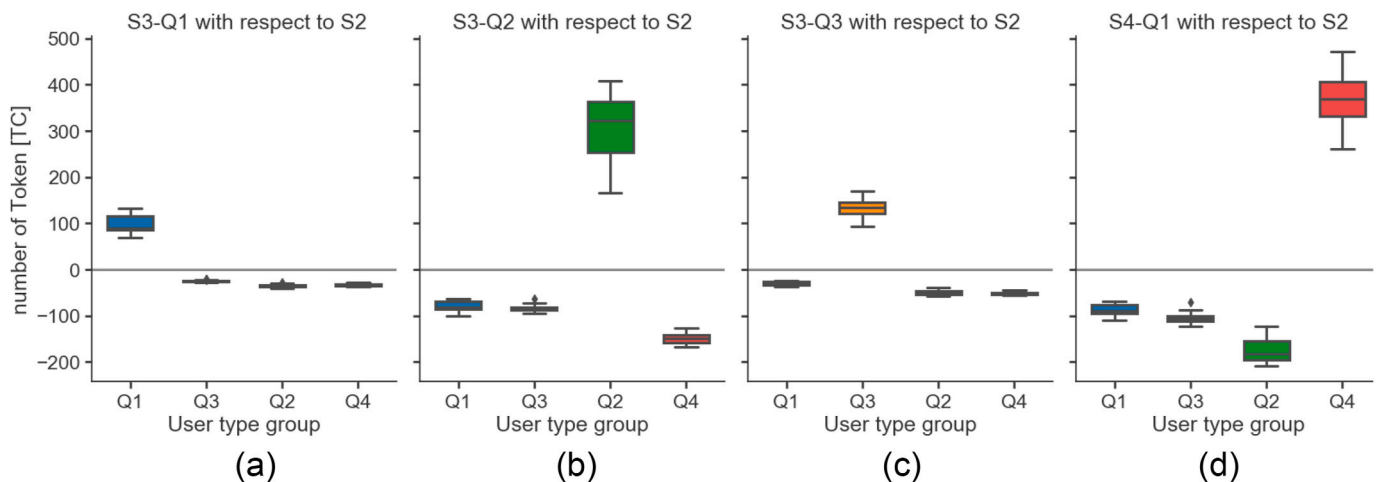


Fig. 9. Difference in the Net Tokens among the UT group members in: (a) the S3-Q1 scenario with respect to the S2 one; (b) the S3-Q2 scenario with respect to the S2 one; (c) the S3-Q3 scenario with respect to the S2 one; (d) the S3-Q4 scenario with respect to the S2 one.

Table 7

Average Token currency gained by each UT member for the different simulated scenarios. The colors go from red (min) to green (max), according to the TC.

Scenario	Sharing Mechanism	Average Token currency for UT groups			
		Q1	Q2	Q3	Q4
S1	SM1	880	880	880	880
S2	SM2	437	1104	551	1462
S3-Q1	SM3	535	1069	525	1429
S3-Q2	SM3	357	1416	466	1313
S3-Q3	SM3	406	1054	684	1411
S3-Q4	SM3	350	930	447	1830

Table 8

Average NPV for the 25th year achieved by an EC member for different sharing mechanism Scenarios and OF schemes. The NPV is expressed as Token per member. The colors go from red (min) to green (max), according to the TC value.

Scenario	SM	Ownership/financing scheme*	Q1	Q2	Q3	Q4
S1	SM1	Type-B, Type-E	261	261	261	261
S2	SM2	Type-B, Type-E	-15	401	56	625
S3-Q1	SM3	Type-C, Type-D, Type-F	-820	668	328	893
S3-Q2	SM3	Type-C, Type-D, Type-F	235	-220	307	866
S3-Q3	SM3	Type-C, Type-D, Type-F	254	659	-727	882
S3-Q4	SM3	Type-C, Type-D, Type-F	219	581	279	-10

*The Type-A and Type-G ownership/financing schemes are not reported since the annual P_{EC} would be reduced because of the extra cost associated with the specific third party or chosen investment fund.

members contribute to financing the generation assets, for example, by adopting an equal share ownership-financing scheme (Type-B, Type-E) or an individual quota OF scheme (Type-C, Type-D, Type-E). In the case of an EC in which all the members contribute equally to the generation assets, investigated by considering the S1 and S2 scenarios, the Q1 and Q3 UT groups benefit the most from the SM1 algorithm. At the same time, they are highly penalized by the SM2 one, as the Q2 and Q2 UT groups obtain the most from the TC because of their higher energy demand.

The EC scenarios with individual quota OF schemes (S3-Q# scenarios) are difficult to generalize. This work investigates a simplified situation where the owners have a fixed individual quota ($Ow_i = CAPEX/25$), while the individual quota of the non-owners is zero ($Ow_i = 0$). The results in Table 8 show that the owner group generally achieves a negative NPV, while the non-owners obtain comparable or higher results than in the S1 and S2 scenarios. The absence of an initial investment leads to a rather high NPV value for the “consumer” members, while, for the owners, it cancels out the benefit of self-consumption priority given by SM3. These results indicate that the SM3 sharing mechanism, even when giving priority to the self-consumption of the owner-members, does not guarantee a fair compensation for providing financing services to the energy community. Therefore, to overcome this issue, the EC should set a different weight for the financing and the self-consumption services in Eq. (13), or give a higher f_i coefficient (Eq. (10))

to the owners to guarantee the proficiency of their investment. Moreover, an EC scenario for SM3, where all the EC members are equal owners, has not been considered here. The reason for this is that such a scenario would correspond exactly to the S2 scenario, as no member would have any self-consumption priority.

The results presented in Table 8 shed some light on what kind of user can benefit from energy community generation asset ownership, under what circumstances, and with what distribution algorithm. Nevertheless, these results cannot be generalized, because of the specific assumptions made about the EC business model, i.e., the lifetime of the incentives, the CAPEX generation asset, the market costs, etc., which are country specific. Moreover, these results could change from EC to EC as they could have a different member composition in each UT group and different EC self-sufficiency levels.

The sharing mechanisms proposed in this work have been investigated in view of their deployment on the operational management platforms of energy communities. Regardless of the size of the energy community and of the actual types of members (other than residential ones), an internal remuneration scheme needs to be set up and it should account for community-owned generation assets, prosumers, and simple consumer members. We thus intentionally explored the case of residential users who have different load profiles (in terms of overall energy demand and temporal matching with the generation supply) and ownership of the generation assets in order to provide a comprehensive

view of a real-life situation in which a community is likely to include users who have quite different energy needs and habits (i.e., different time-of-use patterns), and different financial capabilities to invest (or not) in the generation assets.

Thus, the proposed sharing mechanisms were not tailored for a specific EC setting. They could in fact be applied to any configuration. The only feature that must be maintained is that of a remuneration tariff, which should be attributed to collective shared energy, as is currently the case in Italy, Belgium, Greece, Germany, Portugal and Spain.

5. Conclusion

The main objective of this work has been to investigate how to share profits among energy community (EC) members. An overview of the current energy community frameworks and the possible energy activities has been outlined in the literature review. This work focuses on modeling an energy community within a "Virtual Net-metering" policy scheme, and on a remuneration model, based on the sharing of energy among the community members. A new categorization of energy communities, based on the member ownership of the generation asset(s) and on the adopted financing mechanism, has been introduced.

This work proposes and analyses three different sharing mechanism (SM) algorithms to distribute the EC revenues among the members by means of an internal Token currency and proportionally to the service they provide to the energy community: financing service (SM1), self-consumption service (SM2), and both financing and self-consumption services (SM3).

A synthetic energy community, comprising 100 households supplied by a 100 kWp photovoltaic system, has been simulated to systematically investigate the impact of the adopted sharing mechanism on the EC members. The households were divided into four user type subgroups, with 25 members each, having each subgroup a different energy demand behavior, in terms of total demand and synchronism of the load profile with the PV generation one. The energy community was modeled by simulating one year of dynamic energy exchange at hourly resolution. Energy flows between generation plants and energy community members, through the energy distribution grid, were calculated. The shared energy (SE) of the EC was thus computed. Subsequently, the allocation of the Token currency to each EC member was evaluated, according to the adopted sharing mechanism algorithm. Six scenarios were designed to determine what sharing mechanism was the most suitable for certain user type members, for different PV plant ownership schemes.

The results show that EC members with a low total energy demand (Q1 and Q3) benefit from an equal share of generation asset ownership (Scenario 1) or from being a non-owner member of the EC, as they are rewarded for the self-consumption service (Scenarios S3-Q1 and S3-Q2, respectively). Moreover, achieving synchronism between the load demand and the generation profile is not crucial for these user types, as the amount of achieved TC does not differ substantially between the Q1 and Q3 members. On the other hand, the high energy demand UT members (Q2 and Q4) gain more Token currency when the sharing mechanism awards the self-consumption service (SM2) rather than the ownership one (SM1). In the SM3 sharing mechanism, where both services are rewarded, the members' load demand behavior affects the results to a great extent. In fact, Q2 UT members, who have a poor synchronism between demand and generation, achieve a negative Net Present Value in the 25th year of activity, whenever they are the UT group owners of the PV plant. On the contrary, highly synchronized load demand (Q4)

members are rewarded more in the S3-Q4 scenario, where they are PV owners with respect to the Q2 group, even though their effort to finance the PV system does not allow them to obtain more TC than the other UT groups. Giving the priority of self-consumption of the owners of the generation assets (SM3), accounting for the assets financing service is not enough to fairly distribute the TC among the EC members. Therefore, the financing service should be valorized more than self-consumption by weighing the Token currency assigned for the different activities in different ways.

The key finding of this work is that an energy community has the chance of tailoring the remuneration mechanism to make some users more profitable than others, depending on their load characteristics. Choosing the right sharing mechanism is therefore a matter of determining *a priori* and within the energy community what a fair mechanism is. In principle, by resorting to the investigated mechanisms, it would be possible to either maximize the profits for those who finance the energy community, those who maximize the overall amount of shared energy, or those who are simply vulnerable users (i.e., with a low energy demand and no financial resources to invest into renewable plants).

Other important aspects that deserve further investigation concern how the different EC compositions (number of members in the UT groups) and/or a different EC self-sufficiency level impact the results presented here. Furthermore, another important research question we consider worthy of being addressed is: what is the best balance between owner and non-owner members within an energy community supplied by fluctuating RES?

Credit author statement

Minuto, FD: Conceptualization, Methodology, Software, Data curation, Investigation, Visualization, Writing - original draft, Writing - review & editing. **Lanzini, A:** Conceptualization, Methodology, Software, Writing - original draft, Supervision, Writing - review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Details on the annual energy demand of the EC members broken down according to the Italian ToU tariff are reported in Figure A-1; the cumulative hourly load demand of the 100 simulated household members is shown for a week in Figure A-2; the simulation of a week's PV generation profile, broken down according to the module orientation, is shown in Figure A-3.

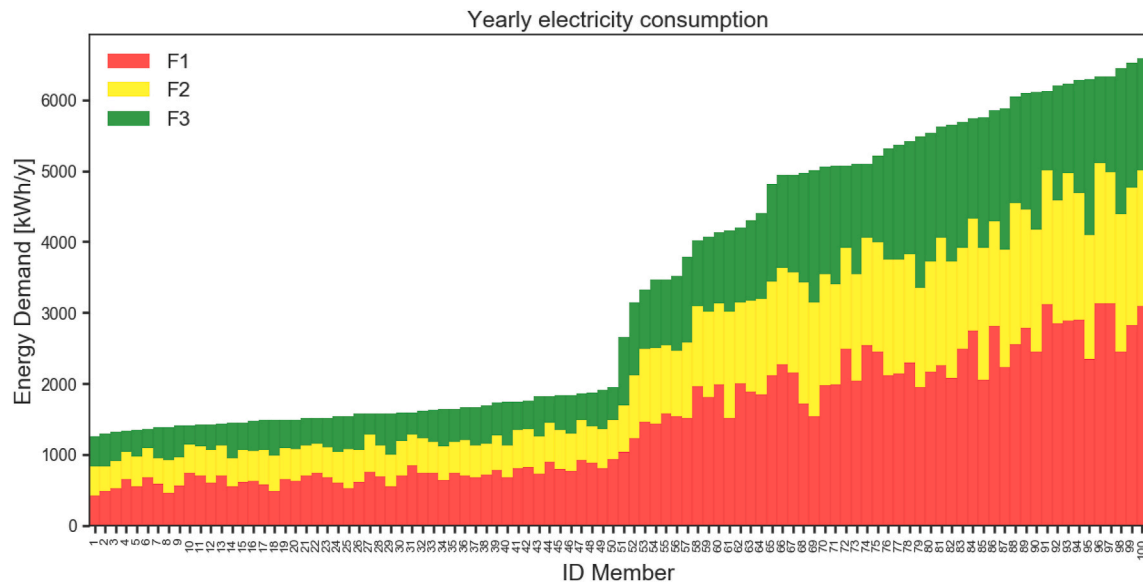


Figure A-1. Annual energy demand of EC members broken down according to the Italian ToU tariff: F1: central hours of the weekdays (8AM-8PM); F2: early mornings (7–8AM), evenings (8–12PM) and Saturdays (F2); F3: Sundays and night hours (12PM-7AM).

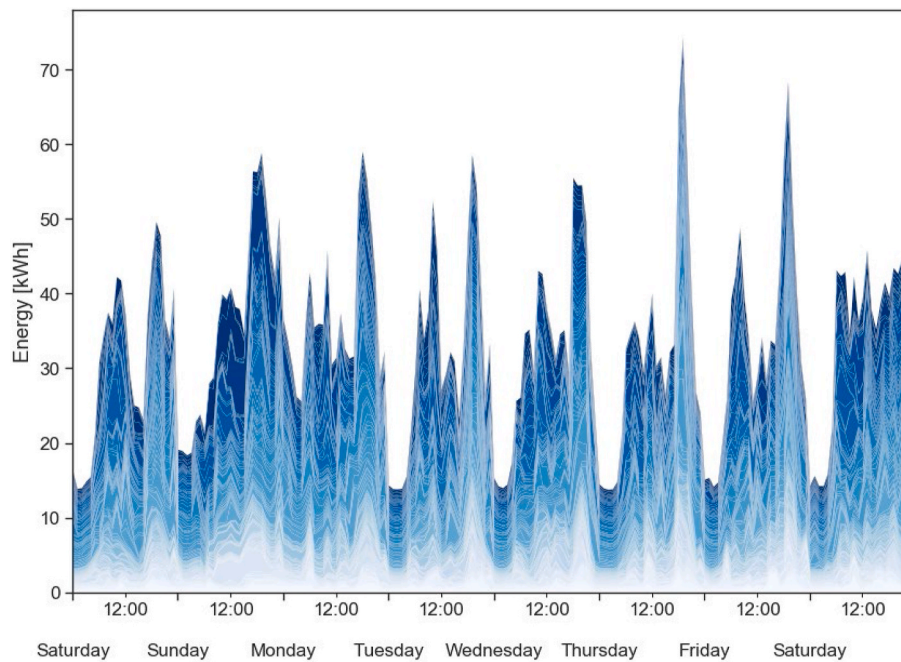


Figure A-2. Example of one week of a simulated household aggregated demand curve (100 EC members) with an hourly resolution; each shade of blue represents the demand of a different user; the cumulated profile is the overall energy demand of the community.

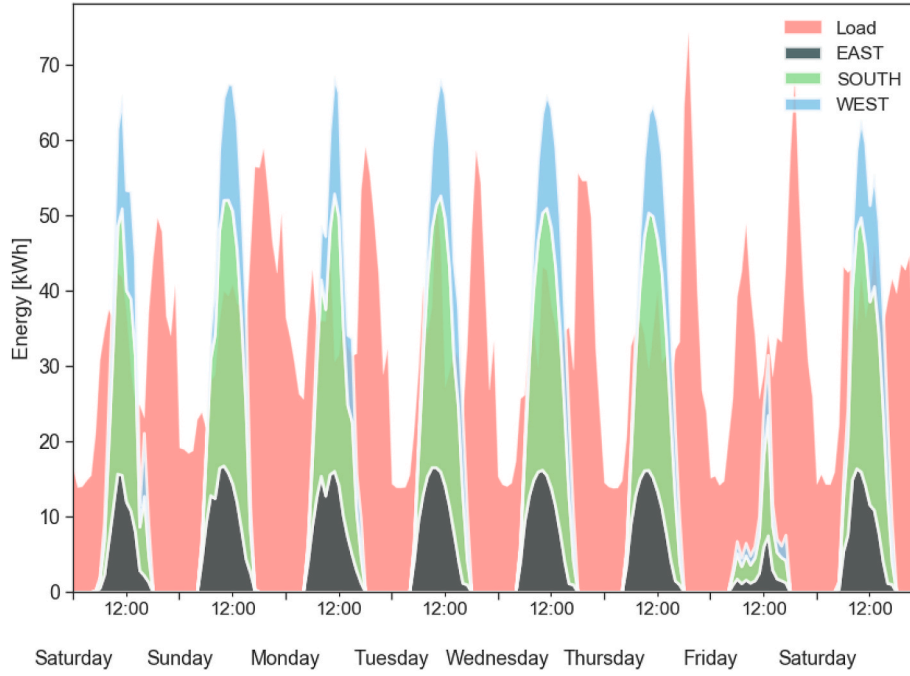


Figure A-3. Stack plot of one week of simulated PV generation for different rooftop orientations for an hourly resolution; the total EC load profile is also plotted for comparison purposes.

Appendix B

The implementation algorithm of the SM2 sharing mechanism is detailed hereafter. The shared energy of the members, SE_i , is calculated by means of the following cyclic algorithm:

1. At the j -th cycle, the lowest energy demand value $L_{min}^j(t)$ is identified among all the users with a greater energy demand than zero.
2. The Partial Shared Energy amount is computed for the j cycle (PSE_j) as:

$$PSE_j(t) = L_{min}^j(t) \cdot M$$

where M is the number of users with a greater or equal energy demand to $L_{min}(t)$.

3. It is verified that the partial share energy, PSE_j , obtained in the previous step, does not exceed the amount of shared energy that has to be allocated, $SE^j(t)$:

$$\text{where } SE^j(t) = SE(t) - SE_i^{j-1}(t) \cdot M$$

$$\text{with } SE_i^{j=0}(t) = 0$$

In this case, the shared energy of the i -th user at step j , $SE_i^j(t)$, is calculated as:

$$SE_i^j(t) = \begin{cases} \frac{SE^j(t)}{M}, & PSE_j(t) \geq SE^j(t) \\ L_{min}^j(t), & PSE_j(t) < SE^j(t) \end{cases}$$

4. Steps 1 to 3 are repeated, when $PSE_{j+1} < SE^{j+1}(t)$, and the user's demand is updated in the subsequent cycle ($j + 1$):

$$L_i^{j+1}(t) = L_i^j(t) - SE_i^j(t)$$

5. Finally, the total shared energy for the i -th user at time step t is calculated as:

$$SE_i = \sum_{j=1}^n SE_i^j(t)$$

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